

# Metahuman Theatre

## Teaching Photogrammetry and Mocap as a Performing Arts Process

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### ABSTRACT

I have been designing a real-time Metahuman animation pipeline in our Media Arts, Design and Technology (MADTech) program at NCSU that emphasizes motion capture as a performing arts process to create dramatic scenes with digital doubles. Students developed a 3D self-portrait using photogrammetry in Epic Games Reality Capture software and Mesh to Metahuman plugin in Unreal Engine, and then performed their double using our inertial mocap suits and control rigs in Autodesk Maya and UE5 sequencer. Facial performance was captured using real-time software like Epic's Livelink Face app and Metahuman Animator, completing a virtual production process that emphasized embodied acting and improvisation over scripts and storyboarding. By framing the Metahuman as a performing object like a puppet and adapting design frameworks like Disney's 12 Principles of Animation to performance capture, students learned a technical animation process in the context of a performing arts approach. This produced aesthetics that are different and arguably more performative than the traditional keyframed approach taught in many animation programs.

### CCS CONCEPTS

• Animation; • Applied computing; • Education; Modeling; • Media arts;

### KEYWORDS

Metahumans, Photogrammetry, Performance Capture, Digital Doubles

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## 1 INTRODUCTION

I started developing our Metahuman photogrammetry and performance capture pipeline to make it easy for upper-level students in our NCSU Media Arts, Design and Technology (MADTech) program to start using real-time technologies as an alternative to more traditional keyframing methods taught in our other 3D animation

courses. My fall 2023 Creative Technology Studio course had students use Reality Capture photogrammetry software to develop Metahuman self-portraits, and then perform their digital doubles with a Noitom Perception Neuron inertial motion capture suit in Unreal Engine 5. While the Metahuman animation pipeline is a technical process that requires knowledge of a variety of hardware and software, many tedious tasks like modeling and rigging are automated so that the students can spend more time on higher-level scene design and character performance. The real-time and embodied capture process of using Epic's Livelink plugins and apps allowed me to frame the pipeline according to live performing arts practices like puppetry and physical acting. I further developed my performative approach in a spring 2024 Motion Capture course by using facial capture tools like Rokoko's Headrig with the Metahuman Animator app to have students create monologues. Then I taught a streamlined version of the full photogrammetry and performance capture pipeline to students in my Hyper-Reality graduate studio course over the last few weeks of the spring 2024 semester.

The photogrammetry process is fun for students because they teamed up in small groups to shoot each other on video, with one person sitting in a chair to stare at a point on the wall and the other using any cell phone to get full coverage of their partner's face and ears. They then imported the video to Reality Capture to extract the frames and the free software stitched a 3D self-portrait that was textured and exported to Autodesk Maya for final cleanup. They then imported their head sculptures into Unreal Engine 5.2 to use the Mesh to Metahuman plugin and the online Metahuman Creator toolset. Self-portraits have a long history in the arts because the reference model is always available to the artist, and throughout the Metahuman creation process we critiqued the quality of their digital double and employed artistic techniques for enhancing their Metahuman's likeness through asymmetries. There is also a level of personal investment in the outcome when you create a representation of yourself, and our graduate students became engaged with the philosophical questions of developing a meta-self.

In my fall 2023 undergraduate studio, students designed one-location scenes that would allow for improvisation around a theme, such as cooking a loaf of bread. We did not create written scripts or storyboards, but instead I had them focus on their acting in the capture volume. My approach was influenced by performing arts principles from *Theatre Anthropology* [Barba, 2003] and *Moment Work* from the Tectonic Theatre Project [Kaufman, 2018], where moments of improvisation are refined to build a dramatic scene. I have also been adapting Disney's 12 Principles of Animation [Johnston, 1981] to performance capture by drawing from the fact that Disney animators studied live performers since the 1930s. In my graduate studio, Metahumans were compared to puppetry concepts like Edward Gordon Craig's Uber-Marionette [Craig, 1907], and we used

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**Table 1: Metadata in tabular format.**

|                   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Summary           | Students use photogrammetry and performance capture to animate Metahuman doubles in dramatic scenes.                |
| Learning Outcomes | Students learn how to create an expressive Metahuman by applying performing arts principles to real-time animation. |
| Classification(s) | Animation, Modeling, Media arts.                                                                                    |
| Audience          | Upper-level students in animation, media arts, game design, or interactive media.                                   |
| Dependencies      | Most software-apps are free and run on a games-level computer or smart phone, but some mocap hardware is costly.    |
| Prerequisites     | Upper-level students in the College of Design who have taken foundation courses.                                    |
| Strengths         | Students enjoy performing creative scenes with a Metahuman that looks like them.                                    |
| Weaknesses        | Students may struggle with the mocap clean-up process or secondary prop animation.                                  |
| Variants          | Students can customize their characters and scenes with many free assets.                                           |
| Assessment        | Students are assessed through weekly critiques and a final animation with a poster.                                 |

VR magician Curtis Hickman’s definition of Hyper-Reality [Hickman, 2023] to design scenes that simulated impossible experiences, such as a student talking to her deceased grandmother through her Metahuman. Software demos and discussion of research on hyper-real characters like *Digi-Doug* from Digital Domain [Roble, 2018] were supplemented with lectures that analyzed the expressive quality of Metahuman performances like *Hellblade 2: Senua’s Saga* [IGN, 2023].

Facial capture was added late in my fall 2023 studio course by having students use iPhones to record directly into Unreal Engine using the Livelink Face app, and then we upgraded to the Metahuman Animator for capturing monologues in the spring. We were initially recording body and face in separate sessions, which I likened to the early days of cinema when sound had to be recorded separately from film. But newer hardware and software releases streamlined the performance process by the spring semester when we started using Rokoko’s Headrig to capture the face and body together and transitioned from using Maya to UE5 for mocap data cleanup. This new technology and streamlined process not only sped up production, but it also dramatically improved the results. While the inertial mocap suits may be cost prohibitive for some programs, we have also been experimenting with the MoveAI smartphone app which allows for one-minute captures at a reasonable monthly subscription cost. This may be preferable in a class with many students because it takes a significant amount of time to suit up and calibrate the inertial suits, and there are fitting and hygiene considerations when sharing equipment.

## 2 COURSE MATERIALS

For student work see our *Metahuman Highlight Reel 2023-24* (along with other tutorials): [https://youtu.be/eDNvmBantDU?si=\\$5WuUhkYNmwHaD6ni](https://youtu.be/eDNvmBantDU?si=$5WuUhkYNmwHaD6ni) Click for an undergraduate poster showing the process: <https://tophermaraffi.com/wp-content/uploads/2024/02/laurametahumanposter.jpg> Here is a graduate poster showing the latest pipeline for a cultural heritage project: [https://tophermaraffi.com/wp-content/uploads/2024/05/rebecca\\_pyun\\_s2024.jpg](https://tophermaraffi.com/wp-content/uploads/2024/05/rebecca_pyun_s2024.jpg) And here is a slide of my adaptation of Disney’s Principles of Animation for performance capture: <https://tophermaraffi.files.wordpress.com/2024/05/disneys12principlesmocapmaraffi.jpg> Please contact me for the course syllabus.

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## 3 RESULTS AND DISCUSSION

The results are different than student work done through a traditional 3D animation pipeline of storyboarding and pose to pose keyframing of a pre-planned narrative. Using photogrammetry increased student “creative investment” [Bennett, 2023] in the Metahuman development process, and added another dimension of meaning to the character models. In addition, the real-time feedback of the performance capture process encouraged student creativity through experimentation. There are expressive ticks that were improvised by each student during the capture sessions that enhanced character appeal, whether for drama or comedy. For example, Laura Reeve’s *Sunday Morning* animation shows a slice-of-life moment of baking bread, but the performer’s unique ticks were only captured because she improvised the scene live, and they create the character’s appeal. There are many examples where the aesthetics go beyond the technical quality of the animation to an expressive liveness that crosses the uncanny valley due to performing arts principles applied in the virtual production process.

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